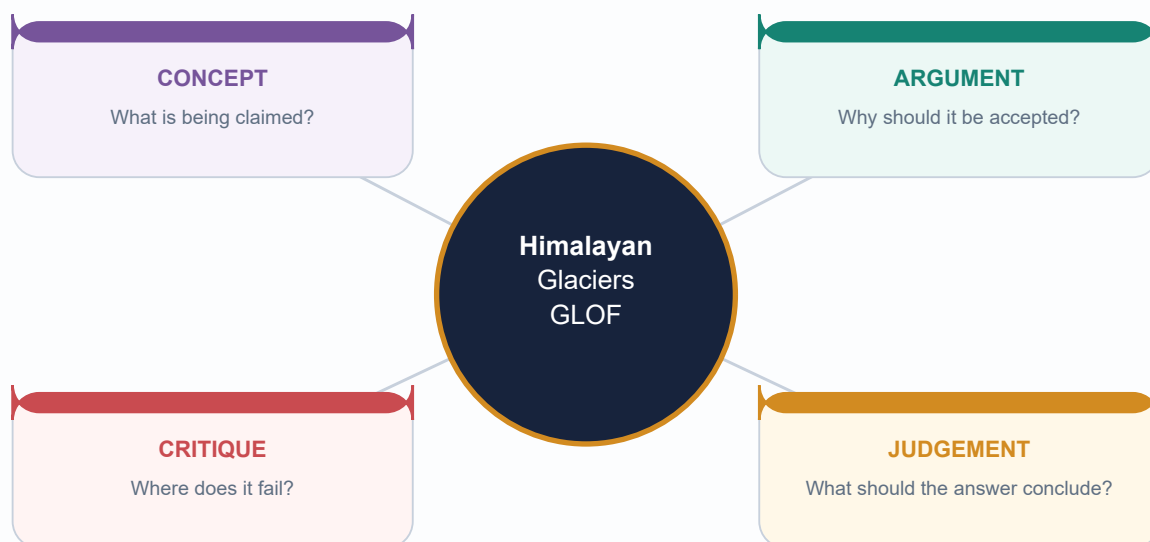


UPSC SOLVED PRACTICE WORKBOOK

06 Himalayan Glaciers and GLOF - Premium Solved Workbook

Geography 06 | Export date: 16 August 2026

Original MCQ keys rotate strictly A -> B -> C -> D. Historical PYQ options retain their original paper order.
Every model Mains answer follows claim -> named evidence/example -> analysis -> qualification and ends
Why this earns marks.



Facts from official sources | Inferences clearly marked | No fabricated data

HIGH UPSC RELEVANCE

HIGH

Workbook protocol: evidence, keys and self-check

GS-I | Prelims |
bounded GS-III
Geography

INTRO & ORIGIN

Use the paper stems first, then compare your answer with the model. Direct Geography PYQs are separated from adjacent ice-map/ice-mass routes.

KEY DATA

Practice type	Source status	How to use
2020/2023 GS-I	Stem verified in local official PDFs	Write to word limit; compare evidence and qualification
2019/2020/2021 Prelims	Stem verified; official older key not held locally	Key labelled INFERRED with elimination/confidence
Original MCQs	Package authored	Strict A-B-C-D key rotation; attempt before solution
Original Mains	Package authored	Use examiner-rubric self-check after model

STATIC THEORY

- [FACT] Local-paper provenance: csp-p1.pdf (2019 Prelims), CSP_2020_GS_Paper-1.pdf, QP-CSP-21-GeneralStudiesPaper-I-121021.pdf, Gen_St_P1.pdf (2020 GS-I), QP-CSM-21-GENSTUDIESPAPER-I-110122.pdf and QP-CSM-23-GENERAL-STUDIES-PAPER-I-180923.pdf.

- [RULE] An unavailable official key is not silently reconstructed. The inferred label states the logic and confidence.

- [SELF-CHECK] For every Mains response: demand fidelity; thesis; 2-3 or more named evidence units proportional to marks; explanation; qualification; conclusion; labelled sketch where it helps.

MUST-KNOW FACTS

1. Do PYQs timed.
2. Do not memorise inferred keys without the elimination logic.
3. Keep direct Geography and adjacent polar/disaster routes distinct.

UPSC TRAPS

WRONG: Treating a coaching answer as an official key.

CORRECT: Only an official local key is official; otherwise use the inference label.

MAINS ANGLE

Use the 'Why this earns marks' note as a marking checklist, not as decorative commentary.

STUDY LINK

Main package Sections 17-20; local PYQ ledgers and official scans.

HIGH

Solved Prelims PYQ - 2019 Q38: glacier-river source matching

GS-I | Prelims |
bounded GS-III
Geography

INTRO & ORIGIN

Official local stem from csp-p1.pdf, page 17. The official answer key is not held locally.

STATIC THEORY

- **Question (verbatim English content):** Consider the following pairs: 1. Bandarpunch - Yamuna; 2. Bara Shigri - Chenab; 3. Milam - Mandakini; 4. Siachen - Nubra; 5. Zemu - Manas. Which of the pairs given above are correctly matched? A. 1, 2 and 4 only B. 2 and 3 only C. 3 only D. 1, 2 and 3.

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: A (1, 2 and 4 only), high confidence.** Bandarpunch is a Yamuna-source cue; Bara Shigri is linked to Chenab; Siachen is linked to Nubra. Milam is associated with Gori Ganga, not Mandakini; Zemu is linked to Teesta, not Manas.

- **Elimination lesson:** map questions often hide two correct broad associations among two wrong tributary/source pairs. Draw the river family, not merely glacier names.

MUST-KNOW FACTS

1. Do not call the inferred key official.
2. Milam-Gori Ganga and Zemu-Teesta are the discriminators.

UPSC TRAPS

WRONG: Assuming all Himalayan glacier-river pairs are familiar trunk rivers.

CORRECT: Check the immediate river system and tributary hierarchy.

MAINS ANGLE

Map-based elimination: identify three certain pairs before deciding the option.

STUDY LINK

Main final map safeguards; Topic 05 drainage.

HIGH**Solved Prelims PYQ - 2020 Q96: Siachen location**

GS-I | Prelims |
bounded GS-III
Geography

INTRO & ORIGIN

Official local stem from CSP_2020_GS_Paper-1.pdf, page 41. The paper's official key is not held locally.

STATIC THEORY

- **Question:** Siachen Glacier is situated to the: A. East of Aksai Chin B. East of Leh C. North of Gilgit D. North of Nubra Valley.

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: D, high confidence.** North of Nubra Valley is the defensible locational relation in the official stem. The question tests map placement rather than glacier physics.

- **Elimination lesson:** retain careful wording. Siachen is an India-administered reference in the eastern Karakoram; avoid turning a map answer into a legal-boundary claim.

MUST-KNOW FACTS

1. Siachen is a map cue, not evidence that all Karakoram glaciers are Indian.
2. Do not conflate Nubra Valley, Gilgit and Aksai Chin relations.

UPSC TRAPS

WRONG: Treating broad regional memory as precise orientation.

CORRECT: Use a simple Karakoram-Ladakh-Nubra sketch.

MAINS ANGLE

Adjacent map route only; do not substitute it for mass-balance explanation.

STUDY LINK

Himalayan regions schematic; geography boundary safeguards.

HIGH

Solved Prelims PYQ - 2021 Q62: water stores and polar ice

GS-I | Prelims |
bounded GS-III
Geography

INTRO & ORIGIN

Official local stem from QP-CSP-21-GeneralStudiesPaper-I-121021.pdf, page 29. This is an adjacent ice-mass route.

STATIC THEORY

- **Question:** With reference to water on planet Earth, consider: 1. The amount of water in rivers and lakes is more than groundwater. 2. The amount of water in polar ice caps and glaciers is more than groundwater. Which is/are correct? A. 1 only B. 2 only C. Both 1 and 2 D. Neither 1 nor 2.

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: B (2 only), high confidence.** Rivers/lakes are a much smaller store than groundwater, whereas polar ice caps/glaciers exceed groundwater as a global water-store category.

- **Concept link:** Store size is not the same as availability or annual renewable flow; tie this to the grounded/floating rule only when sea-level mechanism is asked.

MUST-KNOW FACTS

1. Global stock versus accessible flow is a common Prelims switch.
2. This is global water distribution, not a claim about Indian river discharge.

UPSC TRAPS

WRONG: Equating large water stock with readily available freshwater.

CORRECT: Separate volume, accessibility, turnover and location.

MAINS ANGLE

Adjacent ice-cap route; revise with grounded/floating and Arctic/Antarctic distinctions.

STUDY LINK

Main Section 8; 2021 GS-I adjacent question.

HIGH

Solved Mains PYQ - 2020 GS-I Q6: Himalayan glacier melting and water resources

GS-I | Prelims |
bounded GS-III
Geography

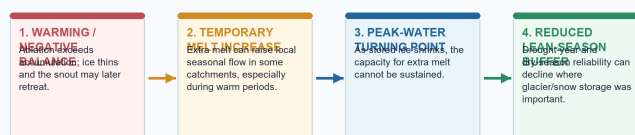
INTRO & ORIGIN

Official local stem from Gen_St_P1.pdf, page 2. Directive: 'How will' requires mechanisms and consequences, not only a climate-change list.

WATER-RESPONSE ANSWER VISUAL

Glacier retreat and the phased seasonal-flow response

The key water-security issue is timing and buffering, not a universal claim that rivers 'immediately dry'.



Qualification: monsoon rainfall dominates much annual flow in many basins; glacier and snow contributions can still matter

Original deterministic schematic | Conceptual and not to scale

Use this phase diagram as a small labelled value-addition, not as a deterministic calendar.

Original deterministic schematic; conceptual, not to scale and not factual measurement.

STATIC THEORY

- **Question:** How will the melting of Himalayan glaciers have a far-reaching impact on the water resources of India? (Answer in 150 words)

- **Model answer:** [CLAIM] Himalayan glacier loss is far-reaching mainly because it changes the seasonality and reliability of water in high mountain headwaters, not because every downstream river will immediately lose all flow. [EVIDENCE] Gomukh on the Gangotri Glacier is the Bhagirathi headstream cue; NCPOR field monitoring and the ISRO Himalayan Glacier Inventory Atlas show why assessment must be glacier/basin specific. [ANALYSIS] Sustained negative mass balance can initially increase local melt contribution, but, after ice volume is reduced, the meltwater reservoir provides less lean-season and drought-year buffering. This can affect irrigation timing, hydropower planning, sediment management, ecosystems and downstream storage. Retreat may also leave moraine-dammed lakes, adding a hazard dimension. [QUALIFICATION] Monsoon rain supplies much annual flow in many basins; seasonal snow, groundwater, catchment elevation, debris cover and glacier fraction vary. Thus the strongest conclusion concerns timing and resilience, not a universal annual-discharge collapse. [CONCLUSION] Basin monitoring, demand/storage planning and risk-aware infrastructure should accompany emissions reduction. **Why this earns marks:** direct thesis, named India/science evidence, phased hydrological mechanism, multiple resource impacts and a necessary basin qualification.

- **Outstanding-answer checklist:** Define the change in one line; draw ELA-snout or phase arrow; use 2-3 named evidence cues; distinguish annual flow from lean-season/drought buffering; end conditionally.

MUST-KNOW FACTS

1. Claim-evidence-analysis-qualification is visible.
2. Water resources includes timing, quality/sediment, ecosystems and infrastructure - not volume alone.

UPSC TRAPS

WRONG: Saying 'all Himalayan rivers will dry'.

CORRECT: Use phased, basin/season-qualified hydrology.

MAINS ANGLE

10-mark allocation: 20-word thesis; 100-word body with mechanism/evidence; 25-word qualification/conclusion.

STUDY LINK

Main Sections 2, 3, 10 and 11.

HIGH

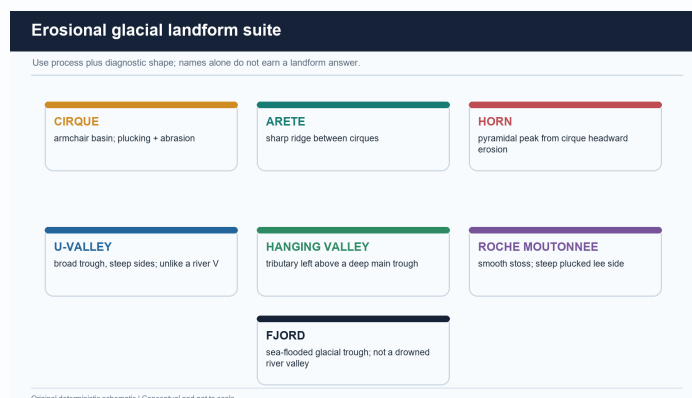
Solved Mains PYQ - 2023 GS-I Q6: fjord formation and picturesque character

GS-I | Prelims |
bounded GS-III
Geography

INTRO & ORIGIN

Official local stem from QP-CSM-23-GENERAL-STUDIES-PAPER-I-180923.pdf, page 2. It asks two linked dimensions: formation and character.

FJORD ANSWER VISUAL



Use the U-trough -> marine inundation sequence and contrast it with a ria.

Original deterministic schematic; conceptual, not to scale and not factual measurement.

STATIC THEORY

- **Question:** How are the fjords formed? Why do they constitute some of the most picturesque areas of the world? (Answer in 150 words)

- **Model answer:** [CLAIM] Fjords are deep, narrow sea inlets that occupy former glacial troughs; their scenery is a direct expression of glacial overdeepening and later marine flooding. [EVIDENCE] The glacial geomorphology sequence in Majid/Leong links valley-glacier abrasion and plucking to a broad U-shaped trough, while hanging valleys provide a familiar waterfall cue; Norway is a classic regional example. [ANALYSIS] Thick moving valley ice widens and deepens the main valley, often below later sea level. When the glacier retreats and the sea enters, steep rock walls enclose deep water. High relief, waterfalls from hanging tributaries, waterfalls/cascades and irregular shorelines explain the picturesque character. [QUALIFICATION] A fjord is not a ria: a ria is a drowned river valley and lacks the diagnostic glacial trough origin. [CONCLUSION] The same process that produces the trough supplies the form's scenic contrast. **Why this earns marks:** it answers both limbs, gives named evidence, explains the two-stage sequence and uses the key comparator.

- **Outstanding-answer checklist:** Write 'glacial trough + marine inundation'; include U-shape/hanging-valley evidence; add ria contrast; avoid calling it a delta or river valley.

MUST-KNOW FACTS

1. Fjord is erosional trough plus submergence.
2. Picturesque character must be physically explained.

UPSC TRAPS

WRONG: Calling a fjord a drowned river valley.

CORRECT: Use ria for drowned river valley; fjord for drowned glacial trough.

MAINS ANGLE

One cross-section earns more than a list of scenic adjectives.

STUDY LINK

Main Sections 5-6; coastal Topic 10.

HIGH

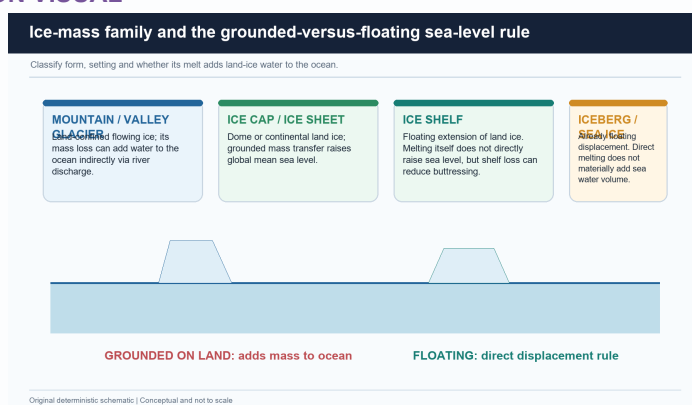
Solved Mains PYQ - 2021 GS-I Q15: Arctic ice and Antarctic glaciers (adjacent)

GS-I | Prelims |
bounded GS-III
Geography

INTRO & ORIGIN

Official local stem from QP-CSM-21-GENSTUDIESPAPER-I-110122.pdf, page 4. Direct owner: polar-climate topic; retained here for ice-mass precision.

ICE-MASS COMPARISON VISUAL



Use this only for an ice-family comparison; it is adjacent to the Himalayan owner.

Original deterministic schematic; conceptual, not to scale and not factual measurement.

STATIC THEORY

- **Question:** How do the melting of the Arctic ice and glaciers of the Antarctic differently affect the weather patterns and human activities on the Earth? Explain. (Answer in 250 words)

- **Model answer:** [CLAIM] The key difference is between mainly floating Arctic sea ice and Antarctic land-based ice-sheet/glacier mass, although both can influence climate through albedo and ocean-atmosphere interactions. [EVIDENCE] IPCC AR6 identifies ice-albedo feedback; the grounded/floating distinction explains why Antarctic grounded ice loss directly adds ocean mass while sea-ice melt follows the displacement rule. [ANALYSIS] Arctic sea-ice loss exposes dark ocean, increasing absorbed energy and potentially altering regional atmospheric/oceanic gradients; consequences for particular mid-latitude weather patterns require confidence-qualified language. Antarctic grounded-ice discharge primarily raises global mean sea level, affecting deltas, low islands and coastal cities, while shelf loss can reduce buttressing of inland ice. [QUALIFICATION] The Arctic also includes Greenland's grounded ice, and Antarctic sea ice is floating; neither region can be reduced to one uniform process. [CONCLUSION] Distinguish ice form, feedback and pathway before discussing human impacts. **Why this earns marks:** it answers the comparison directly, uses IPCC/physical evidence, avoids overclaiming weather causation and gives the grounded/floating qualification.

- **Why it appears here:** It tests the same ice-family mistake that can weaken Himalayan answers; it is not a substitute for the 2020 Himalayan-water PYQ.

MUST-KNOW FACTS

1. Floating sea ice and grounded ice have distinct direct sea-level effects.
2. Weather linkage needs confidence-qualified wording.

UPSC TRAPS

WRONG: All Arctic ice is floating.

CORRECT: Greenland is grounded land ice; distinguish components.

MAINS ANGLE

Adjacent comparison only; use Topic 25 for full polar coverage.

STUDY LINK

Main Section 8; Geography Basic 25.

HIGH

Hard original MCQs 1-8: mechanics, landforms and ice family

GS-I | Prelims |
bounded GS-III
Geography

INTRO & ORIGIN

Attempt without the answer sheet. Keys are A, B, C, D, A, B, C, D.

STATIC THEORY

- **Q1.** Which observation is strongest evidence of a negative annual glacier mass balance?
 - A. Annual ablation exceeds annual accumulation over the stated balance period
 - B. The glacier contains a cirque
 - C. A river is perennial downstream
 - D. Snow lies on a shaded slope in one month

- **Q2.** Which statement best explains why a snout can retreat while glacier ice moves down-valley?
 - A. Retreat reverses glacier flow direction
 - B. Ablation at the terminus exceeds forward ice delivery
 - C. The ELA becomes the snout
 - D. Moraine becomes an ice shelf

- **Q3.** Which combination correctly identifies direct ice deposition versus meltwater deposition?
 - A. Moraine sorted; outwash unsorted
 - B. Esker unsorted till; drumlin river gravel
 - C. Moraine/till generally unsorted; outwash relatively sorted
 - D. Fjord till; ria outwash

- **Q4.** A roche moutonnée is most reliably recognised by:
 - A. a lake enclosed by terminal moraine
 - B. a ridge of sorted sand/gravel
 - C. a V-shaped river valley at a coast
 - D. a smoother abraded stoss side and a steeper plucked lee side

- **Q5.** Which statement about an ice shelf is correct?
 - A. It is a floating extension of land ice and can buttress grounded ice behind it
 - B. It is an upland dome smaller than an ice sheet
 - C. It directly raises sea level when it melts because it was grounded
 - D. It is another name for a cirque glacier

- **Q6.** Which is the clearest difference between climatic snowline and glacier snout?
 - A. Snowline is always below the snout
 - B. Snowline concerns persistent snow conditions; snout is the lower ice terminus
 - C. Snout is a balance line; snowline is a moraine
 - D. Both are fixed regional elevation contours

- **Q7.** Which inference about Himalayan debris cover is safest?
 - A. Debris always increases melting
 - B. Debris always protects ice from melting
 - C. Thin debris may enhance melt while thick debris may insulate; local geometry matters
 - D. Debris cover makes glacier mass balance unmeasurable

- **Q8.** Which form is correctly matched to its origin?
 - A. Ria - drowned glacial trough
 - B. Esker - terminal ice front ridge of till
 - C. Drumlin - sea-flooded U valley
 - D. Fjord - marine-inundated glacial trough

MUST-KNOW FACTS

1. Question order/key sequence is checked before export.
2. All distractors are process or category errors, not trivia.

UPSC TRAPS

WRONG: Peeking at the rotation instead of reasoning.
CORRECT: Rotation is an audit control, not a shortcut.

MAINS ANGLE

If an answer is wrong, write the false premise in your error log.

STUDY LINK

Solutions next; then risk and map MCQs.

HIGH**Hard original MCQs 1-8: solutions**

GS-I | Prelims |
 bounded GS-III
 Geography

STATIC THEORY

- **Q1 A.** Balance is the annual/period gain-minus-loss condition. A cirque or a perennial river does not determine the balance.
- **Q2 B.** The ice may move down-valley, but terminus loss can outrun its delivery.
- **Q3 C.** Ice drops mixed till; flowing meltwater sorts material by hydraulic action.
- **Q4 D.** The asymmetric abraded stoss / plucked lee profile is the diagnostic cue.
- **Q5 A.** Floating shelves do not directly add their displaced volume on melt, but their buttressing matters for grounded discharge.
- **Q6 B.** Persistent snow conditions/balance references differ from a lower dynamic ice terminus.
- **Q7 C.** Debris-thickness effect is conditional; that nuance defeats both absolutes.
- **Q8 D.** Fjord = marine-flooded glacial trough. The other options deliberately swap origins.

MUST-KNOW FACTS

1. Keys: A B C D A B C D.
2. Each explanation names the disqualifying mechanism in the distractors.

UPSC TRAPS

WRONG: Memorising 'till unsorted' without agent.
CORRECT: Remember: direct ice drop versus flowing meltwater.

MAINS ANGLE

Translate two wrong answers into a 20-word landform comparison.

STUDY LINK

Final register notes: erosion/deposition and ice family.

HIGH**Hard original MCQs 9-16: Himalayan water and GLOF chain**

GS-I | Prelims |
 bounded GS-III
 Geography

INTRO & ORIGIN

Continue the same original-key rotation: A, B, C, D, A, B, C, D.

STATIC THEORY

- **Q9.** Which statement most accurately describes Himalayan glacier contribution to river flow?
 - A. It can be especially important for lean-season/drought buffering in some glacierised catchments, while monsoon dominates much annual flow in many basins
 - B. It supplies identical annual shares to all Himalayan rivers
 - C. It is irrelevant wherever monsoon rain falls
 - D. It guarantees constant discharge
- **Q10.** The 'peak-water' idea in a shrinking glacier catchment most directly means:
 - A. annual rainfall has permanently ended
 - B. extra melt contribution may rise temporarily before a smaller ice store reduces it later
 - C. the snout reaches the sea
 - D. every lake has breached
- **Q11.** Which is correctly classified as a predisposition rather than necessarily an immediate GLOF trigger?
 - A. A displacement wave overtopping a dam
 - B. An upstream lake outburst entering another lake
 - C. Unconsolidated moraine with seepage paths and exposed downstream assets
 - D. Sudden breach incision through the dam
- **Q12.** Which sequence most plausibly describes a GLOF cascade?
 - A. Fjord -> glacier -> ria -> sea-level rise
 - B. Snowline -> terminal moraine -> drumming -> esker
 - C. Warming -> guaranteed breach -> exact forecast
 - D. Lake/dam setting -> trigger/impulse -> overtopping or internal erosion -> breach -> water-sediment flood
- **Q13.** Why is a moraine-dammed lake not automatically a disaster?
 - A. Risk also requires dam/slope condition, a possible trigger and downstream exposure/vulnerability
 - B. Moraine cannot impound water
 - C. GLOFs occur only from ice dams
 - D. Lake expansion proves breach timing
- **Q14.** What is the most defensible use of South Lhonak-Teesta in an answer?
 - A. Proof that every retreating lake will fail
 - B. A bounded multihazard case: lake growth, analysed lateral-moraine/impulse-breach sequence and downstream exposure, with background trigger uncertainty retained
 - C. Evidence that earthquakes are the only GLOF cause
 - D. A source for uncited casualty/infrastructure figures
- **Q15.** Which intervention most directly connects lake monitoring to life safety?
 - A. Publishing a one-time inventory only
 - B. Measuring no variables until after a breach
 - C. Thresholds, interoperable warning messages, evacuation routes/drills and last-mile communication
 - D. Treating every lake as equally dangerous
- **Q16.** In a high mountain hydropower corridor, which is the best risk-reduction principle?
 - A. Ignore sediment because water is the only hazard
 - B. Rely on a single historical flood value without review
 - C. Put all response responsibility on households
 - D. Test GLOF/inundation/sediment scenarios, siting, design, access and warning together

MUST-KNOW FACTS

1. Water timing and hazard chain use different causal diagrams.
2. South Lhonak must retain an uncertainty boundary.

UPSC TRAPS**WRONG:** Calling a lake map a GLOF prediction.**CORRECT:** It is screening/context; prediction and risk reduction need further evidence and preparedness.**MAINS ANGLE**

For a GLOF MCQ, identify whether the option is about hazard, risk, trigger or response.

STUDY LINK

Solutions next; then remedial traps and answer-writing questions.

HIGH**Hard original MCQs 9-16: solutions**

GS-I | Prelims |
bounded GS-III
Geography

STATIC THEORY

- **Q9 A.** This retains both lean-season significance and monsoon-dominant annual-flow qualification.
- **Q10 B.** It is a phase logic: temporary extra melt can be followed by declining meltwater buffering.
- **Q11 C.** Moraine/seepage/exposure create susceptibility; a displacement wave or breach is an event stage.
- **Q12 D.** The option keeps physical stages in causal order without pretending prediction certainty.
- **Q13 A.** Hazard becomes risk through failure likelihood plus exposure/vulnerability, not lake presence alone.
- **Q14 B.** This uses the case analytically while preserving uncertainty and avoids unsupported figures.
- **Q15 C.** Monitoring works through thresholds, communication and evacuation readiness.
- **Q16 D.** The corridor must be analysed as water-sediment-infrastructure cascade, not a single design number.

MUST-KNOW FACTS

1. Keys: A B C D A B C D.
2. The solution explanations deliberately distinguish causal stage from intervention.

UPSC TRAPS**WRONG:** Choosing the option with the most agencies.**CORRECT:** Choose the one with a defensible physical-to-social causal chain.**MAINS ANGLE**

Write a four-box GLOF chain from memory before doing the remedial set.

STUDY LINK

Main Sections 12-15; Disaster Management
Topic 10.

HIGH**Remedial original MCQs 17-24: common UPSC traps**

GS-I | Prelims |
bounded GS-III
Geography

INTRO & ORIGIN

These are error-correction questions. Original keys continue A, B, C, D, A, B, C, D.

STATIC THEORY

- **Q17.** Which statement corrects the error 'all Himalayan rivers are glacier-fed in the same way'?

- A. Glacier/snow, monsoon rain and groundwater contributions vary by basin, elevation and season
- B. Only glaciers feed perennial rivers
- C. All Peninsular rivers are dry all year
- D. Annual flow alone measures water security

- **Q18.** A student writes 'thin debris insulates and thick debris accelerates melting'. The best correction is:

- A. debris effect is always zero
- B. thin debris can enhance melt while thick debris may insulate, subject to setting
- C. all debris converts ice to a fjord
- D. debris only affects floating ice

- **Q19.** Which statement is most accurate about South Lhonak?

- A. A single settled trigger has been officially proved beyond debate
- B. Lake expansion alone fixes a failure date
- C. Use a bounded chain: moraine-dammed lake context, analysed lateral-moraine/impulse/overtopping-breach sequence, and uncertain background contributions
- D. The case eliminates the need for lake monitoring elsewhere

- **Q20.** Which is a valid distinction between a GLOF and a landslide?

- A. Both are simply monsoon floods
- B. A GLOF cannot carry sediment
- C. A landslide always occurs under a glacier
- D. A landslide is gravity-driven mass movement; it may trigger a GLOF, which is rapid release from a glacial lake

- **Q21.** Which answer opening best fits 'Critically examine GLOF risk in the Himalaya'?

- A. 'GLOF risk is a lake-dam-and-exposure problem; retreat can raise predisposition but site triggers and valley development determine outcomes.'
- B. 'Glaciers are beautiful natural features.'
- C. 'All climate disasters have one cause.'
- D. 'The answer is awareness.'

- **Q22.** Which pairing is incorrect?

- A. Milam - Gori Ganga
- B. Zemu - Manas
- C. Bara Shigri - Chenab
- D. Siachen - Nubra

- **Q23.** A fjord answer earns additional marks when it:

- A. calls it a delta
- B. treats picturesque character as a tourist statistic
- C. links glacial trough excavation, marine inundation, steep relief and hanging-valley waterfalls, then distinguishes a ria
- D. says tides cut a V-valley into a fjord

- **Q24.** Which closing is strongest for a Himalayan glacier-change answer?

- A. 'No action is possible because climate is global.'
- B. 'Every basin needs the same dam.'
- C. 'All historical numbers prove current trends.'
- D. 'Adaptation should manage seasonal uncertainty, lake/slope cascades and exposure through basin-specific monitoring and risk-aware design.'

MUST-KNOW FACTS

1. A wrong option usually violates agent, stage, scale or evidence standard.
2. Use correction MCQs to repair answer language.

UPSC TRAPS

WRONG: Thinking the longest option is automatically correct.

CORRECT: Test the causal claim and qualifiers word by word.

MAINS ANGLE

Convert Q21 and Q24 into an introduction and conclusion template.

STUDY LINK

Solutions immediately follow.

HIGH**Remedial original MCQs 17-24: solutions**

GS-I | Prelims |
bounded GS-III
Geography

STATIC THEORY

- Q17 A. Basin/season/source heterogeneity corrects the false uniformity claim.
- Q18 B. The debris effect depends on thickness and setting; both universal claims fail.
- Q19 C. It is the evidence-disciplined case formulation and retains uncertainty.
- Q20 D. A landslide can be a GLOF trigger but is a distinct gravity mass movement.
- Q21 A. It gives a thesis, causal structure and critical qualification from the first line.
- Q22 B. Zemu is associated with Teesta, not Manas; the other pairs are the map eliminators.
- Q23 C. Formation plus geomorphic scenic mechanism plus ria comparison answers every part.
- Q24 D. It reaches a conditional, mechanism-matched adaptation verdict.

MUST-KNOW FACTS

1. Keys: A B C D A B C D.
2. Across original workbook MCQs 1-24 the key sequence is A-B-C-D repeated six times.

UPSC TRAPS

WRONG: Using the rotation as a substitute for knowing the topic.

CORRECT: The rotation validates the set; it cannot validate a guessed mechanism.

MAINS ANGLE

Review every incorrect answer with one causal diagram or comparison sentence.

STUDY LINK

Final original Mains models below.

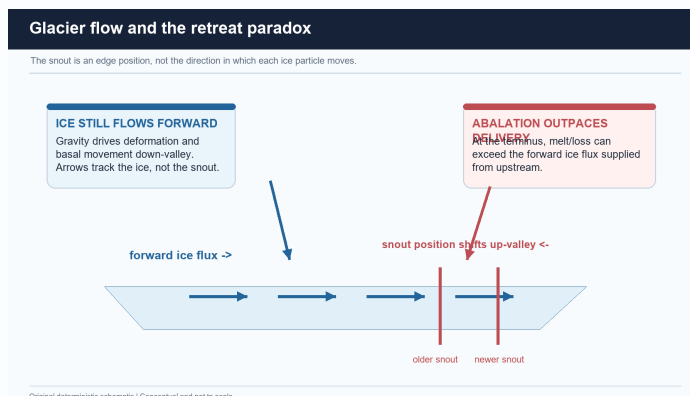
HIGH**Original solved 10-marker: flow, retreat and water reliability**

GS-I | Prelims |
bounded GS-III
Geography

INTRO & ORIGIN

Answer in 150 words: Explain why a retreating Himalayan glacier can still flow downslope. Assess one water-security implication.

10-MARK PROCESS VISUAL



A small arrow diagram can prevent the common retreat/backward-flow error.

Original deterministic schematic; conceptual, not to scale and not factual measurement.

STATIC THEORY

- **Model answer:** [CLAIM] A retreating glacier still flows down-valley because retreat is the movement of its terminus, not reversal of ice motion. [EVIDENCE] The ELA-snout budget and Gangotri/Gomukh valley-glacier setting show that gravity-driven ice delivery continues while lower-tongue ablation can exceed it. [ANALYSIS] When melt/loss at the terminus is faster than upstream ice flux, the visible snout shifts up-valley. Sustained negative balance also reduces stored ice; after any temporary extra-melt phase, dry-season/drought-year buffering in a glacierised headwater can weaken. [QUALIFICATION] Monsoon rain, seasonal snow, groundwater and catchment geometry mean this is not an automatic annual-flow collapse for every river. [CONCLUSION] The operational concern is seasonal reliability, so monitor balance, snow and discharge together. **Why this earns marks:** it corrects the core misconception, uses a named Himalayan cue, explains hydrology and limits the claim.

- **Examiner self-check:** Did you distinguish particle motion from terminus position? Did you state one realistic, qualified water consequence?

MUST-KNOW FACTS

1. Claim → evidence → analysis → qualification visibly structured.
2. One labelled arrow diagram is enough value addition.

UPSC TRAPS

WRONG: Spending 100 words on generic global warming.

CORRECT: Give the mechanics first; climate forcing is context, not the whole answer.

MAINS ANGLE

10-marker target: 3 evidence/cue units, one qualification and a direct conclusion.

STUDY LINK

Main Sections 2-3 and 11.

HIGH

Original solved 15-marker: GLOF risk and valley-floor development

GS-I | Prelims |
bounded GS-III
Geography

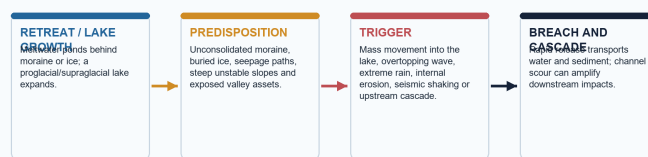
INTRO & ORIGIN

Answer in 250 words: Discuss GLOF risk as an interaction between cryospheric change and valley-floor development. Use South Lhonak-Teesta critically.

15-MARK GLOF VISUAL

Moraine-dammed lake and the GLOF trigger-breach-cascade chain

Separate predisposition, trigger, dam failure and downstream risk.



Risk = hazard x exposure x vulnerability. A lake's presence is not itself a forecast of failure.

Original deterministic schematic | Conceptual and not to scale

Use the physical chain to organise a GS-I/GS-III boundary answer.

Original deterministic schematic; conceptual, not to scale and not factual measurement.

STATIC THEORY

- **Model answer:** [CLAIM] GLOF risk is an interaction between an evolving lake-dam system and the exposure created in steep downstream valleys; it cannot be reduced to glacier retreat or one trigger. [EVIDENCE] CWC describes ice/moraine/natural depressions impounding water and warns that unstable impoundment can release large volumes quickly. South Lhonak was a moraine-dammed lake that expanded as its glacier retreated; post-event analyses examine lateral-moraine movement, impulse/overtopping and breach. [ANALYSIS] Retreat can create lake space and expose weak moraine; a mass movement, intense rain, seepage/internal erosion or cascade can trigger overtopping/breach. Water and sediment then scour narrow Teesta corridors where roads, bridges and hydropower can amplify exposure. [QUALIFICATION] Lake expansion is a predisposition, not a dateable forecast; relative roles of rainfall, permafrost, seepage and other conditions remain under research. [CONCLUSION] Combine remote/field assessment, thresholds and alerts, drills, hazard zoning, design-flood/sediment review and site-specific lake treatment. **Why this earns marks:** it integrates physical mechanism, a carefully bounded case, exposure and matched action without claiming false certainty.

- **Examiner self-check:** Are hazard, risk, trigger and warning used correctly? Does each response target a chain stage?

MUST-KNOW FACTS

1. Use South Lhonak with an uncertainty boundary.
2. Do not quote death tolls or loss figures without source/date.

UPSC TRAPS

WRONG: Adding 'awareness' without causal linkage.**CORRECT:** Show how communication/evacuation reduces vulnerability after warning.

MAINS ANGLE

15-marker target: 4-6 named evidence/case/institution units, one risk-chain diagram and a critical judgement.

STUDY LINK

Main Sections 12-15; Disaster Management Topic 10.

HIGH

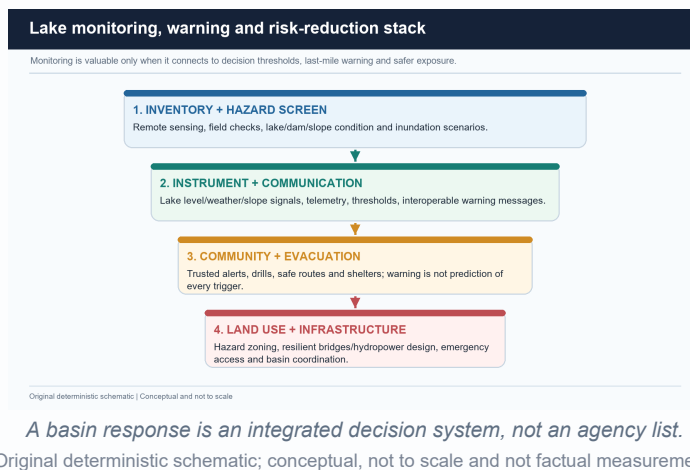
Original solved 20-marker: basin-scale cryosphere adaptation

GS-I | Prelims |
bounded GS-III
Geography

INTRO & ORIGIN

Answer in 250 words: Evaluate Himalayan glacier change as a water, hazard and landscape problem. Frame a basin-scale adaptation response.

20-MARK ADAPTATION STACK



STATIC THEORY

- **Model answer:** [CLAIM] Himalayan glacier change is best evaluated as a coupled water-timing, lake/slope hazard and sediment-landscape problem, whose consequences depend on basin characteristics and exposure. [EVIDENCE] Majid's glacial-process account supplies the mass-balance/landform base; NCPOR/ISRO monitoring illustrates glacier-specific observation; CWC's GLOF material identifies lake/dam uncertainty; South Lhonak-Teesta shows a multihazard corridor cascade. [ANALYSIS] Negative balance shifts ELA/ice volume; early extra melt can give way to reduced lean-season buffering. Retreat exposes moraine/rock, alters drainage and sediment connectivity, and may enlarge lakes that can fail after a trigger. A basin plan should observe mass balance/snow/discharge/lake level/slope condition, map inundation and sediment pathways, set warning thresholds and drills, assess outlet-lowering where feasible, and align zoning, bridge/hydropower design, access and ecosystem flow. [QUALIFICATION] Monsoon dominance, debris cover, altitude/aspect, local geology and data gaps mean one rate or intervention cannot fit all basins; operational emergency doctrine remains with competent disaster authorities. [CONCLUSION] The durable strategy manages uncertainty and exposure while reducing emissions and improving cryosphere knowledge. **Why this earns marks:** it evaluates rather than narrates, joins five named evidence units to mechanisms, qualifies regional variation and produces a coherent basin response.

- **Examiner self-check:** Did you evaluate criteria (water/hazard/landscape), show links, include 5+ evidence units and deliver a conditional conclusion?

MUST-KNOW FACTS

1. Basin scale connects headwater observation to downstream exposure.
2. A 'risk stack' must include monitoring, decisions and land-use/design.

UPSC TRAPS

WRONG: Writing only about lake warning in a water/hazard/landscape question.

CORRECT: Cover all three demanded dimensions before responses.

MAINS ANGLE

20-marker target: 5-8 evidence units, a labelled causal map and a balanced conclusion.

STUDY LINK

Main complete package; final register notes.